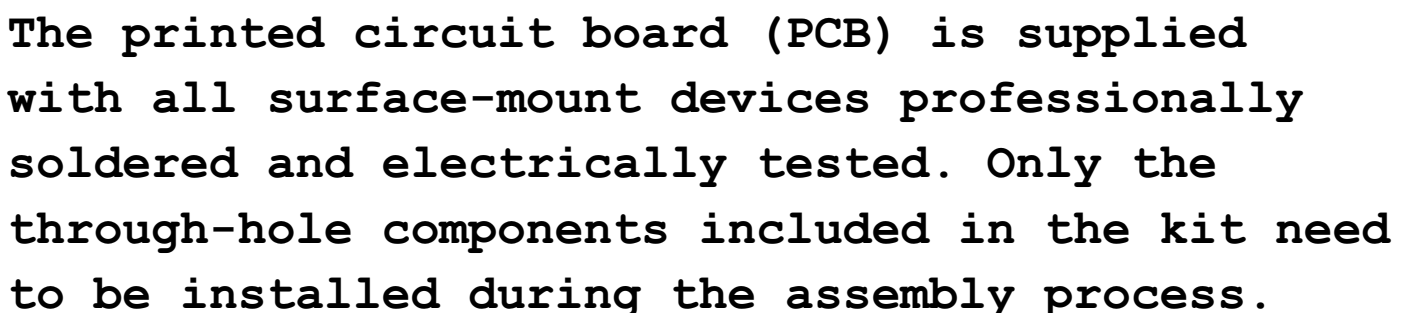


Let's assemble the ESP32 IoT Controller by soldering the female header connectors and the OLED display. Solder the headers first and then the OLED display. Make sure connector sits completely flush against the board, and adjust it before soldering the rest. Take your time to ensure every joint is secure, aiming for a shiny cone rather than a dull solder blob.





Soldering Techniques

- > Keep components flush with the PCB before soldering.
- > Allow enough time for proper heat transfer.
- > Use only enough solder to form a smooth, shiny fillet.
- > Inspect every joint, it must be shiny

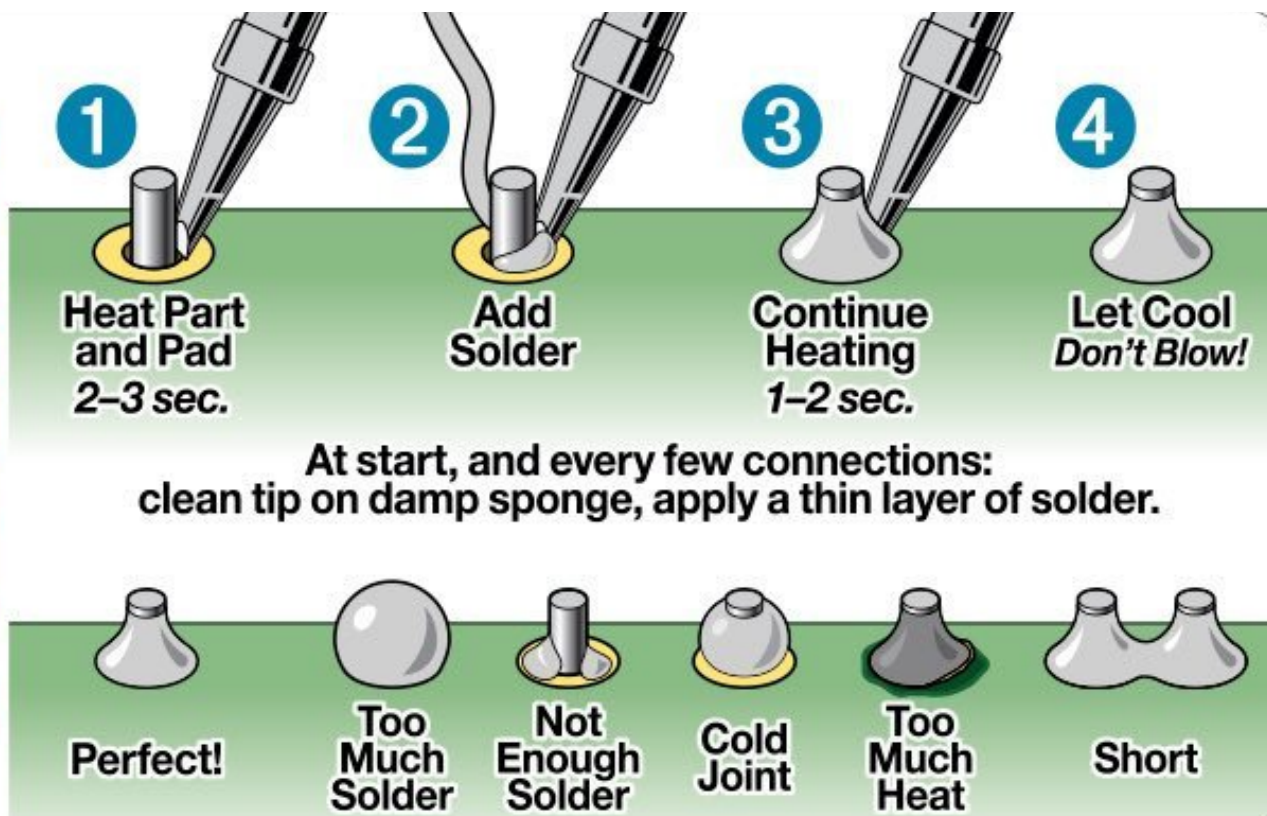
Youtube Videos

- How to solder properly

<https://www.youtube.com/shorts/v0F0BqzjFrc>

- How to solder Header Pins

<https://www.youtube.com/watch?v=E9NIN49iiCc>





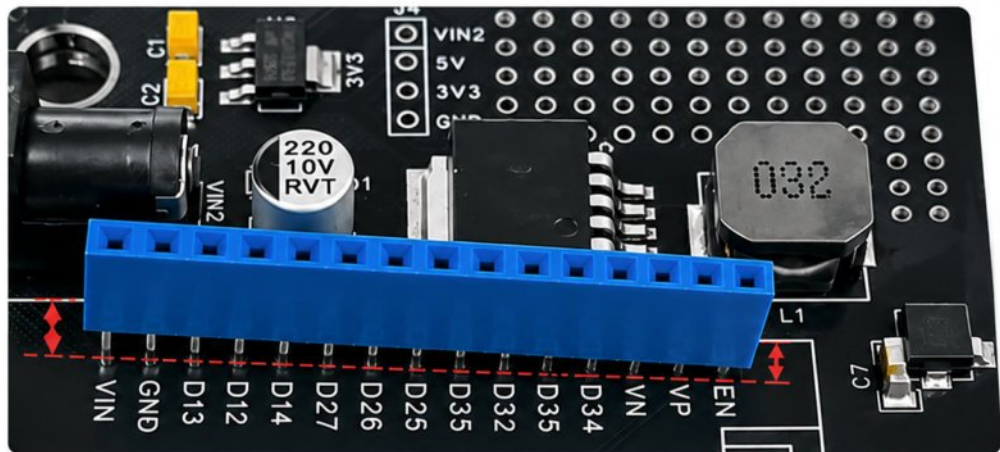
Ensure the connector is flush before soldering

It is essential that the connector is flush with the PCB before soldering. First solder the two outer pins, verify the connector is straight and fully seated, then solder the remaining pins.



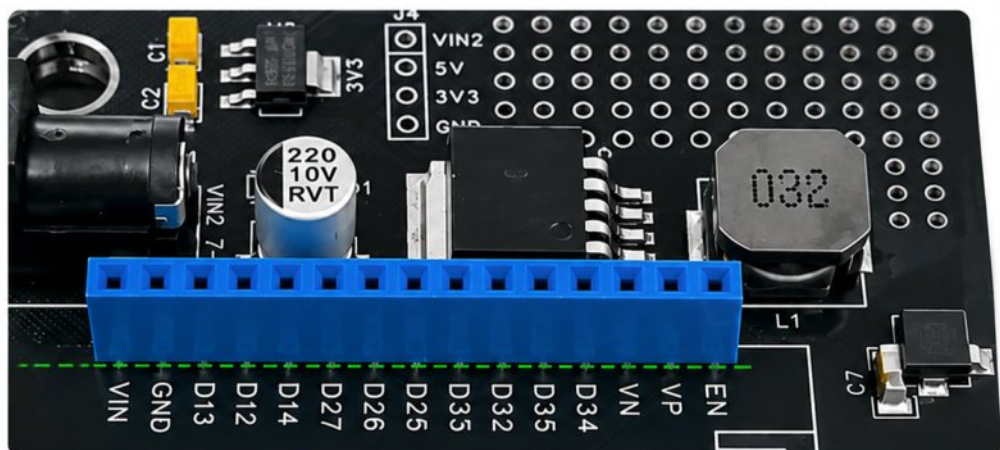
NOT FLUSH & SKEWED

Connector is tilted
and not parallel
to the board



FLUSH & STRAIGHT

Connector is parallel
and flush with
the board



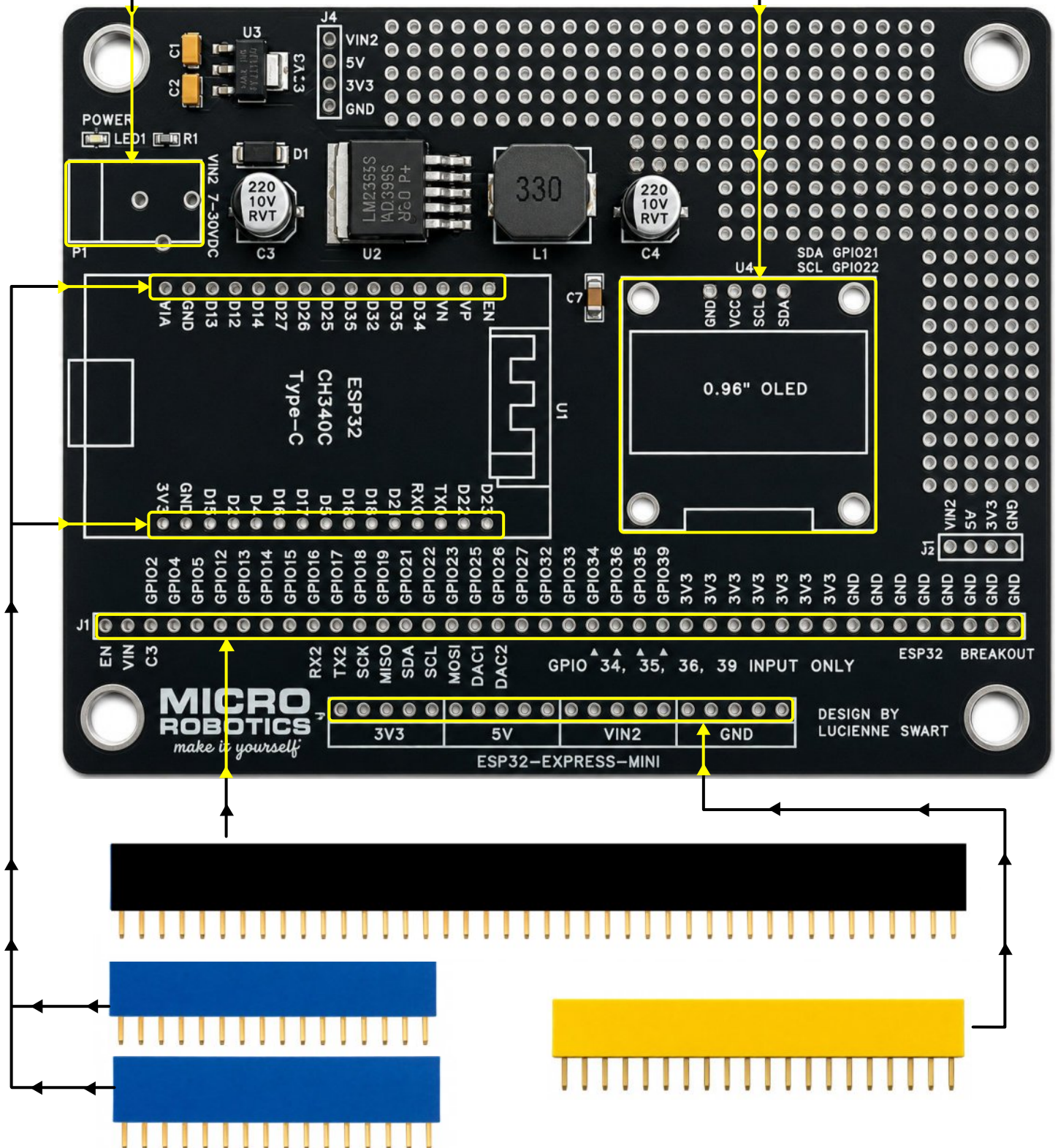
Soldering Components



DC Connector



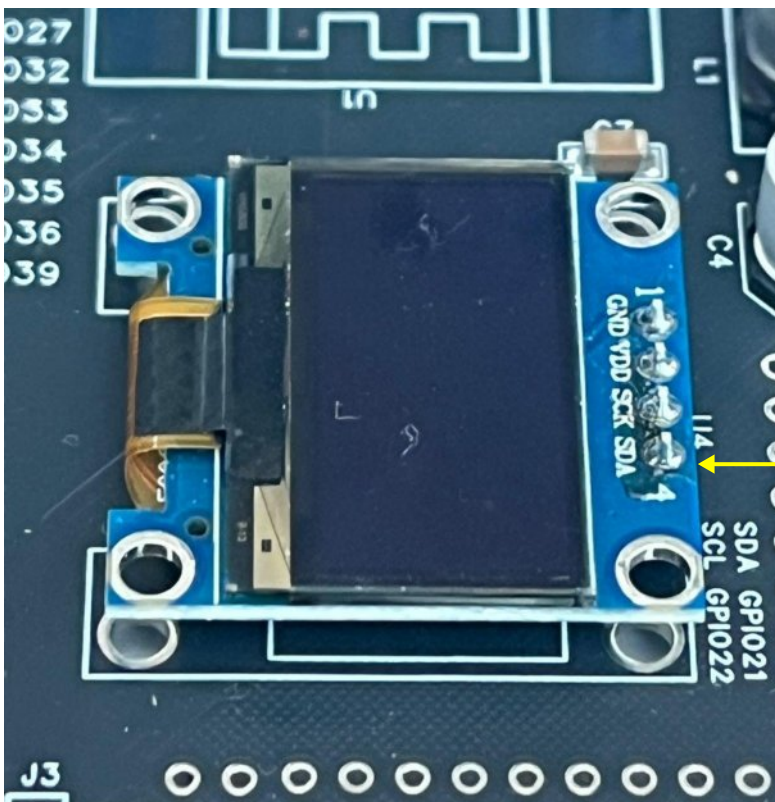
OLED Display



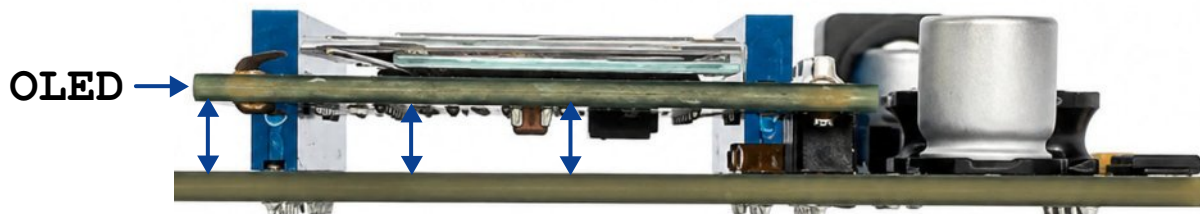


Soldering OLED Display

The OLED module has only four solder connections. Before soldering all four pins, ensure the display is perfectly parallel to the PCB and correctly aligned.



OLED Display

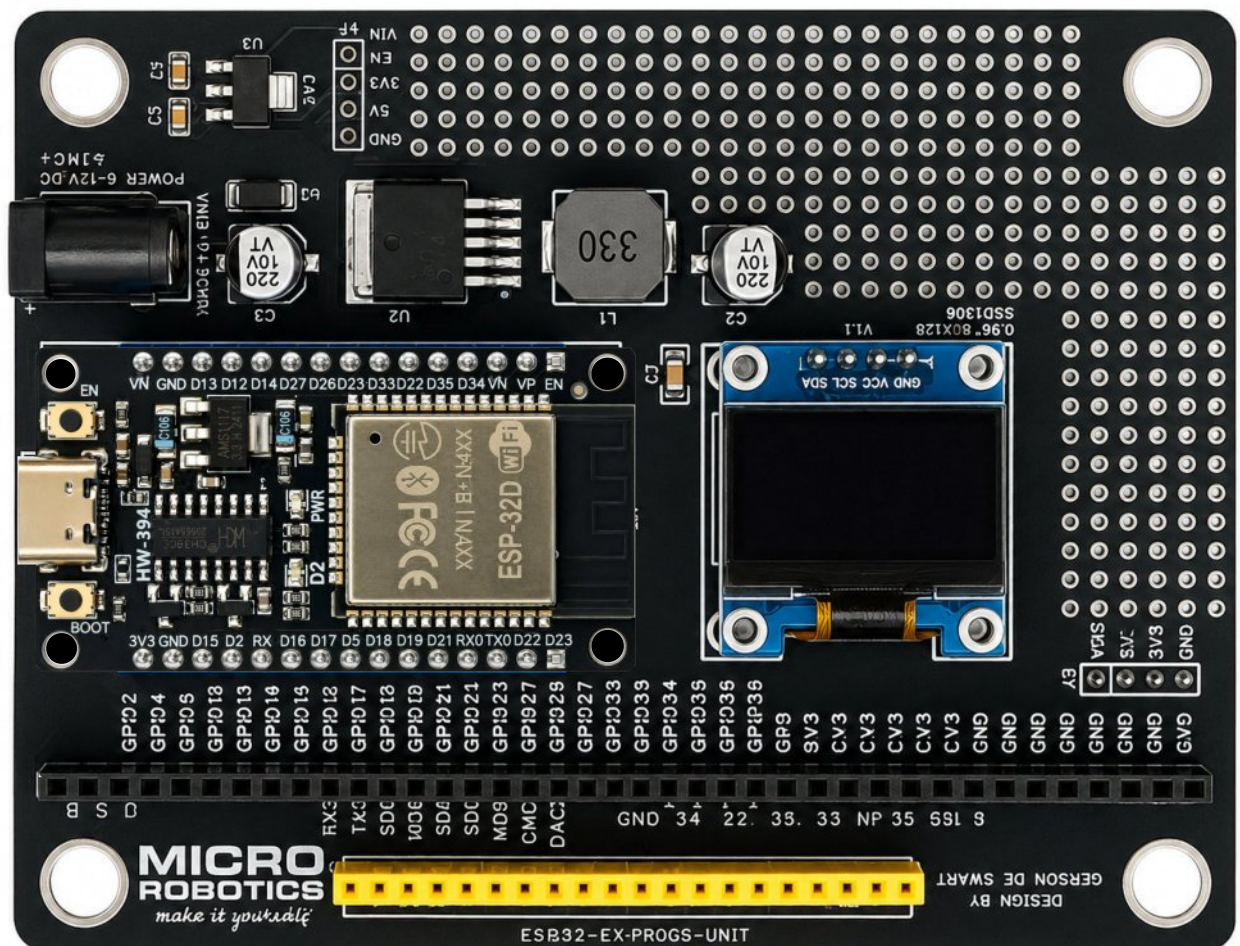


Ensure OLED Display is parallel with PC Board



Installing the ESP32 Module

The board is now fully assembled and ready for the ESP32 module. Ensure the module is correctly oriented before insertion. The USB Type-C connector must align with the outer edge of the PCB. Do not force the module into the headers.



Fully Assembled ESP32 IOT Controller

UP NEXT
Testing the Board

